

MacIntroPsych Study Guide

Welcome to MacIntroPsych! We offer two introductory psychology courses at McMaster: Psych 1X03/1F03 and Psych 1XX3/1FF3. Psych 1X03 and 1XX3 follow a blended learning model. This means you will experience a combination of online learning technology with traditional face-to-face instruction. Psych 1F03 and 1FF3 cover the same content as their counterparts, but do not have an in-person component. They use the same online modules and textbook, but do not have a lecture or tutorial component.

On your way to the weekly quizzes, midterm and final exam, there are several resources available to help you master the curriculum:

Course Handbook: Your course handbook contains valuable information regarding course structure, outlines, and guides for the web modules. It has tons of practice questions, key slides, note guides, concept maps, and a full set of flash cards! It can be purchased at the McMaster Campus Store.

Course Textbook: We use a digital textbook provided through TopHat. You can purchase your access code from the McMaster Campus Store. The textbook contains required readings with testable material for the course.

Avenue to Learn: Your primary course content will be delivered through the Avenue learning management system located at <http://avenue.mcmaster.ca>. Avenue allows you access to weekly web modules, course announcements, discussion forums, and grade records. To access Avenue, use your MacID and password. Below are some of the features of Avenue.

Web Modules: The most unique feature of MacIntroPsych is the way you receive your primary course content—it's all online! You can access the web modules from the library, your room, or anywhere you have an internet connection. The interactive web modules feature audio, video animations, and vivid graphics. Check out the many advanced features that allow you to interact with the content according to your personal learning preferences. Use the navigation tools and integrated search function to move about the module. You can test your knowledge with checkpoints, participate in class polls, and learn more about faculty related research. If you are taking Psych 1X03 or 1XX3, be sure to view the assigned web modules **before** you attend your weekly lecture and tutorial to stay on schedule, get points on the bonus quizzes, and actively participate.

Discussion Boards: More extended topic discussions are available on the Avenue Discussion Board. Join an existing discussion or start a new thread. Our discussion boards are consistently the most active in comparison to any other course on campus, so jump right in with your questions! They are monitored daily by TAs so it is a fast way to get your content related questions answered.

Tutorials: If you are in Psych 1X03 or 1XX3, you will have a tutorial every week. You will join a small tutorial section (capped at 26 students) led by a Teaching Assistant (TA) who is enrolled in or has completed PSYCH 3TT3: Applied Educational Psychology, a course designed specifically to help TAs lead effective tutorials and guide you through MacIntroPsych. Each year, our TAs receive top ratings from students across campus so don't be shy to ask questions. If you think your TA is especially amazing, consider nominating them for the Kathy Steele Award, which honours the top intropsych TA of the year. Your TA will guide discussions, lead activities and demonstrations, and answer any questions you might have. Tutorials are updated each year by feedback from students and TAs.

Lobby Hours: Have a question? Still confused about a specific concept? Need some one-on-one time? Drop by during our office hours to speak with a TA. The times are posted on Avenue. If you have administrative questions please visit Dr. Michelle Cadieux, the course coordinator, or send her an email (intropsych@mcmaster.ca). Her office hours are updated on Avenue weekly.

Taking Notes

Students often wonder (and worry) about how extensive their notes should be. The course handbook provides outlines with key points and slides reproduced from the web modules to guide your own note taking. There really is no substitute for doing this yourself to learn the material. If, however, you can refer to your notes and answer the practice questions that follow the handbook outlines, you should find yourself in good shape for the weekly quizzes and the exam to come.

Writing Tests

With practice questions in the handbook, pre-quizzes and quizzes, you might be wondering “why are there so many tests?!”. The simple answer is that testing has been shown to be the most effective way to learn information in the long term.

Retrieval-Enhanced Learning

Many students likely view testing as a negative necessity of their courses and would prefer to have as few tests as possible. Thinking about testing this way is due to years of experiencing tests as a high-stakes assessment of learning. This is unfortunate given the fact that testing improves learning. Over the past hundred years, research on the characteristics of human learning and memory has demonstrated that practice testing enhances learning and retention of information (e.g., Dunlosky et al., 2013; Roediger & Karpicke, 2006a). Practice testing can take many forms. It can refer to practicing your recall of information by using flashcards, completing practice problems or questions in a textbook, or completing low-stakes tests as part of a course requirement. This principle was the primary motivation for redesigning the MacIntroPsych course to have weekly, low stakes quizzes.

An excellent example of the power of testing memory comes from a study by Roediger and Karpicke (2006b), wherein undergraduate students were presented with short, educationally relevant texts for initial study. Following initial study, students either studied the material again or took a practice test. A final test was taken after a short retention interval (5 minutes) or long retention interval (2 days). After a short retention interval, restudying produced better recall than testing (81% vs. 75%). However, with the long retention interval, testing produced significantly better recall than restudying (68% vs. 54%). Thus, after two days, performance declined 27% for students who restudied the material, but only 7% for students who practiced recall.

Interestingly, providing students with the correct answer feedback after a test enhances the positive effect of testing. With feedback, learners are able to correct errors, and maintain their correct responses. Moreover, taking a test and reviewing feedback can enhance future study sessions. Research shows that when a student takes a test before restudying material, they learn **more** from the restudying session than if they restudy without taking a test beforehand (e.g., Karpicke, 2009). This is called test-potentiated learning.

Why does testing improve retention of information and how can I use it?

Explanations for the positive effects of testing focus on how the act of retrieval affects memory. Specifically, it is suggested that retrieving information leads to an elaboration of memory traces and the creation of additional retrieval paths. Together these changes to memory systems make it more likely that the information will be successfully retrieved again in the future. This suggests that testing is not just an assessment tool, but also an effective learning tool.

As a student in this course, you can take advantage of retrieval enhanced learning each week in preparation for your weekly Quiz. This begins with studying web module content early in the week (e.g., Sunday or Monday). You can then engage in retrieval practice as a form of review after your initial study session. At this point you should be ready to complete the pre-quiz and review the feedback (by Wednesday or Thursday). This gives you the opportunity to take advantage of test-potentiated learning when you review content again before completing your Quiz on Friday. This suggested schedule of studying, and incorporation of retrieval practice will help you learn and retain the course content. Engaging in this process each week enhances your long-term

memory for course content and therefore advances your preparations for the final exam!

Suggested further reading:

1. Dunlosky, et al., (2013). *Psychological Science in the Public Interest*, 14(1), 4-58.
2. Karpicke, J. D. (2009). *Journal of Experimental Psychology: General*, 138, 469–486.
3. Roediger, H. L., & Karpicke, J. D. (2006a). *Psychological Science*, 17, 249–255.
4. Roediger, H. L., & Karpicke, J. D. (2006b). *Perspectives on Psychological Science*, 1, 181–210.

Multiple Choice Questions and Flashcards

Test questions can challenge you to demonstrate your mastery of course content through 3 levels of increasing difficulty:

1. To *remember* psychological terms, concepts, theories, and methods.
2. To *understand* information that has been remembered.
3. To *apply* understood information to novel situations (this is called *transfer* and demonstrates true mastery of learned material).

Let's take a look at an example of a psychological concept you will learn:

Confounding Variable: An extraneous variable whose presence affects the variables being studied so that the results do not reflect the actual relationship between the variables.

Here is a simple question that challenges your ability to *remember* the term:

1. Which of the following best describes a confounding variable?
 - a. A dependent variable whose absence affects the variables being studied so that the results do reflect the actual relationship between variables.
 - b. An independent variable whose absence affects the variables being studied so that the results do not reflect the actual relationship between variables.
 - c. An extraneous variable whose presence affects the variables being studied so that the results do reflect the actual relationship between variables.
 - d. An extraneous variable whose presence affects the variables being studied so that the results do not reflect the actual relationship between variables.

The correct answer is D, and knowing this requires the ability to *remember* (or look up) a word-for-word definition of confounding variable taken directly from the course material

2. If a confounding variable is unnoticed, it can lead a researcher to:
 - a. Make a more accurate connection between an independent and dependent variable.
 - b. Make a more accurate connection between two independent variables.
 - c. Make an erroneous association between an independent and dependent variable.
 - d. Make an erroneous association between several confounding variables.

The correct answer is C, and knowing this requires the ability to *understand* information that has been accurately remembered, because you must be able to recognize a definition of confounding variable that is composed of different words than the original definition.

3. A researcher gives the same IQ test questions to a group of children and discovers that arm length is positively correlated with number of correct responses. He concludes that longer arms lead to increased intelligence. What confounding variable may account for this finding to weaken this conclusion?
 - a. Some children may have been purposely answering test questions incorrectly.

- b. Arm length increases with age, so this study may demonstrate that older children outperform younger children.
- c. IQ is a hypothetical construct that may not accurately reflect intelligence.
- d. IQ test questions may lack validity and reliability.

The correct answer is B, and knowing this requires the ability to *apply* understood information in a novel situation.

Using flashcards effectively can be a solid method for efficiently building your knowledge to prepare for the three types of test questions. On the back of each flashcard is a ready-prepared definition for the term to help you *remember*. When studying, you should try to rework this definition into your own words to help you *understand*. Finally, create an original example that illustrates the term to help you learn to *apply*.

To provide some additional help for effective use of the flashcards, we have included an additional set of cards on the following page. These flashcards will require you to go deeper with the concepts thus increasing your understanding and retention.

Bonus tip: One of the most robust findings in research on the science of learning is the distributed practice effect, which shows the benefits of spreading out study sessions over time. Let's say Thomas and Kristal both spend 4 hours studying the same material for a test. Thomas distributes his practice by studying 1 hour a day for 4 days and Kristal crams all 4 of her study hours in one long study marathon the night before the test. Even with time-on-task equated, Thomas will show significantly improved long-term benefits.

**Describe a movie/television scene
that depicts this concept**

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**Pick two cards and make a
connection between the concepts**

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**Describe this concept without using
any of the key words written on the
flashcard**

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Draw this concept

MAC 
IntroPsych

**How would you explain this concept
to a child?**

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**What is the opposite of this
concept?**

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**What situation in your life has
depicted this concept?**

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**Create a concept map with all of
the cards from a module**

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Campus Resources

The following are some of the resources available to McMaster University students. Read over each description and familiarize yourself with what is available!

Student Accessibility Services

Student Accessibility Services offers various supports for students with disabilities. They work with full-time, part-time, and prospective students. SAS assists with academic and disability-related needs, including learning strategies, note-taking, assistive technologies, test & exam administration, accommodations for courses, groups, and events.

Website: <http://sas.mcmaster.ca/>

Phone: 905-525-9140 ext. 28652

Teletypewriter (TTY): 905-528-4307

Email: sas@mcmaster.ca

Office: McMaster University Student Centre (MUSC) - Basement, Room B107

Student Wellness Centre

The Student Wellness Centre offers medical & health services, personal counselling, and mental health services.

Medical & Health: The Student Wellness Centre provides a wide range of health services for students and will act as their personal health care provider throughout their studies at McMaster. Appointment bookings should be made ahead of time.

Personal Counselling & Mental Health: At some point, almost everyone experiences major concerns that may interfere with their success, happiness, and satisfaction at university. Common concerns are relationships, mood disorders, learning disabilities, body image, anxiety, and depression. The Student Wellness Centre provides experienced counsellors for bookings and emergency appointments.

Website: <http://wellness.mcmaster.ca/>

Phone: 905-525-9140 ext. 27700

Email: wellness@mcmaster.ca

Office: PGCLL 210/201

Emergency First Response Team (EFRT)

The McMaster Students Union Emergency First Response Team (EFRT) is a group of approximately 30 student volunteers who provide emergency medical services to the McMaster Campus, with an average response time of 2-3 minutes. All responders are certified Emergency Medical Responders, with more senior members being trained in International Trauma Life Support (ITLS), and Advanced Medical Life Support (AMLS). The EFRT is available 24/7 throughout the academic year except Thanksgiving, Easter, the Holiday Break, and both Reading Weeks.

Website: <https://www.msumcmaster.ca/services-directory/7-emergency-first-response-team-efrt>

Phone: "88" from any campus phone or 905-522-4135 on any cell phone

The MacPherson Institute for Leadership, Innovation and Excellence in Teaching

This institute is designed to enable success in teaching and learning. Their activities include both general and discipline-specific approaches to the promotion of successful learning in all its forms and contexts.

Website: mi.mcmaster.ca

Phone: 905-525-9140 ext. 24540

Office: Mills Library L504

Academic Advising in PNB

Jennifer Nettleton is the academic advisor for anyone in the PNB department at McMaster. She has a lot of knowledge about upper-year courses and can help you make the best decisions about what courses to take.

Email: pnbadvis@mcmaster.ca

Phone: 905-525-9140 ext. 23005

Office: PC 109

Note: For academic advising for other departments, visit your faculty's website (Social Science, Science, Engineering, etc.) for more information.

A Note from Dr. Joe Kim

We have many talented and passionate members of the Instructional Staff and Development Team that work hard to bring you an outstanding course experience. MacIntroPsych was honoured with the 2010 President's Award for Excellence in Course and Resource Design. Our unique IntroPsych Program has been the topic of academic study and received widespread media attention in the Toronto Star, Globe and Mail, CHCH News, and Maclean's (not to mention Mac's own Daily News). Importantly, the continual development of the MacIntroPsych Blended Learning Environment model is supported by ongoing research. As the Director of the Education & Cognition Laboratory (EdCog Lab), I am actively interested in teaching, learning, and technology from both an academic and a practical perspective. For more information, visit <https://edcog.ca/index.html>

University can sometimes seem like an impersonal and strange place, especially for Level 1 students who are dealing with many adjustments. I hope that in exploring the course resources, you will not forget that there is a real live faculty member responsible for the MacIntroPsych Program—me! I have regular office hours (posted weekly on Avenue) set aside solely to give me a chance to meet and talk with you. If you have a question, comment, complaint, concern, or just want to see and chat with a live faculty member, do come. Many students are reluctant to talk to a faculty member outside of class. Don't give in to the feeling! I have had many great conversations with students that have started off with a supposedly “silly” question.

As a Teaching Professor, my primary responsibilities are teaching and interacting with students; even my area of research interest concerns pedagogy—the formal study of teaching and learning. My goal is to help you understand and appreciate some of the really interesting and important things that we know (or think we know) about human thought and behaviour. In most fields—and as you will see, certainly in psychology—the simplest questions are often the most important and difficult to answer.

One last piece of advice—get involved in the course! MacIntroPsych is a fascinating world waiting to be explored by you! Keep up with the web modules, actively participate in tutorials, join the discussion forums, and watch the PNB talks on MacIntroPsych.com. It really will make all the difference. Following each web module, I would also encourage you to participate in the feedback surveys. Many of the most popular interactive features were suggested by students just like you. I really do read every single comment and they have contributed enormously to minor and major changes made to all aspects of the course, and this includes the very course handbook you hold in your hands.

On behalf of all the wonderful people that work hard on the frontlines and behind the scenes, best of luck and have a great year!

Dr. Joe Kim